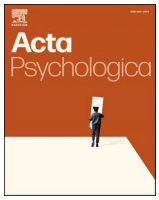
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Online reviews in e-commerce: A bibliometric analysis of consumer decision-making

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ABSTRACT

Over the past two decades of digitalization, Internet technology has reconfigured the information ecosystem of consumer decision-making. Nowadays, consumers actively obtain product information through multi-touch interaction networks, where online reviews have become a core element influencing their purchase decisions. This phenomenon presents enterprises with a dual challenge: to address the issue of processing massive online reviews and to establish a dynamic trust mechanism in virtual consumption scenarios. Based on the Web of Science database literature, this study employs bibliometric analysis to systematically examine the mechanism by which online reviews influence consumers' purchasing decisions. It is found that existing studies can be integrated into seven theoretical dimensions, ranging from the basic layer of sentiment analysis and product feature mining to the behavioral driving layer of online trust construction, and ultimately to the decision-making ecosystem encompassing consumer psychological mechanisms, behavioral modeling, and the identification of false reviews. The study not only establishes an interdisciplinary theoretical framework for analysis but also highlights the crucial role of the balance between real-time data processing technology and privacy protection in e-commerce practice. The paper suggests that we should explore the intelligent analysis system of reviews driven by large language models and the dynamic decision prediction model in cross-cultural scenarios, thereby providing a new research paradigm for precision marketing in the digital economy era.

1. Introduction

The internet technology revolution has continuously reshaped the power dynamics between online platforms and consumers over the past two decades. The social distancing measures imposed by the COVID-19 pandemic acted as a critical catalyst, accelerating the migration of consumer behavior online, with consumption scenarios shifting massively from physical spaces to virtual networks. This has propelled e-commerce systems to achieve generational leaps in service depth (such as round-the-clock intelligent customer service) and scenario breadth (such as virtual scenario marketing and AI outfit recommendations) (Lei et al., 2021). Throughout this process, consumer demands have transcended mere 'commodity transactions' to embrace 'co-created experiences.' The pursuit of emotional resonance, instant feedback, and immersive engagement has compelled platforms to reconstruct user-centric digital communication ecosystems (Kim et al., 2022). The synergy between technology and behavior ultimately catalyzed the awakening of consumer agency: online reviews have supplanted official brand communications as the core nodes within e-commerce knowledge networks, forming a 'digital reputation cloud' spanning all product categories. Driven by the 'stay-at-home economy,' review usage rates and decision-making reliance continue to rise, establishing a new consumption paradigm where 'no decision is made without reviews' (Li et al., 2023; Wu & Liao, 2021).

As online reviews have evolved from "decision references" to "decision foundations", their influence mechanisms have become pivotal to deciphering the underlying logic of digital consumer behavior. Yet existing research remains fragmented and fragmented, lacking a systematic framework. Whilst the e-commerce revolution, empowered by digital technology, has restructured distributed decision-making mechanisms around online reviews—transforming individual decisions into multi-stakeholder cognitive games—this process has also precipitated deep-seated contradictions within the review ecosystem—the 'decentralisation' of information acquisition conflicts with the finite nature of individual cognition. While vast quantities of reviews expand decision-making dimensions, they simultaneously trigger the information

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overload phenomenon where the volume of information exceeds consumers' processing capacity, leading to diminished decision quality or decision fatigue (Duan et al., 2018; Najmi et al., 2015). It is noteworthy that methodological critiques of relevant research approaches (such as surveys and experimental studies) should be grounded in systematic evaluation rather than preconceived judgements. For instance, the accuracy of survey data depends not only on the methodology itself but also on the scientific rigor of questionnaire design and the representativeness of the sampled population (Groves et al., 2011). While experimental studies effectively control variables, their ecological validity requires a comprehensive assessment relative to specific research objectives (Reis & Judd, 2000). Ultimately, the evaluation of research methods should be based on their suitability for the research question and the rigor of their implementation, rather than arbitrary categorization (Creswell & Creswell, 2017).

Research focus has expanded from early emphasis on information quantity to dimensions such as information quality and content inconsistency, with ongoing debate regarding whether recommendation algorithms simplify decision-making or exacerbate cognitive biases. Sentiment polarity—the emotional direction (positive/negative) of reviews—exhibits non-linear effects on purchase intent: its influence depends not only on valence direction but also involves emotional intensity, perceived authenticity, and contextual factors (Zhao & Li, 2021). For instance, moderately negative reviews may enhance credibility through social validation, whereas extremely positive evaluations may trigger authenticity doubts, placing consumers in a credibility-authenticity cognitive dilemma (Chen et al., 2019). Furthermore, trust mechanisms—encompassing platform-based institutional trust and reviewer-based interpersonal trust—emerge as critical moderators of cognitive conflict, with their dynamic construction profoundly shaping information processing pathways. The spatio-temporal diffusion of reviews exhibits a dual 'long-tail-pulse' dichotomy: classic reviews shape product perception through the long-tail effect, while sudden public sentiment can trigger precipitous fluctuations in purchase intent within 24 h, compelling consumers to repeatedly weigh stability against timeliness (Kim & Na, 2021). The intertwined contradictions of information overload, the emotional polarity paradox, and the challenge of trust construction not only reflect the structural impact of the review ecosystem on cognitive systems but also reveal that decision-making mechanisms must shift from 'efficiency-first' to a dual-dimensional balance of 'risk-meaning' (Guo & Zhang, 2024).

Although existing research has examined the influence of review valence, quantity, timeliness and other factors on purchasing decisions, most explorations remain fragmented, lacking a systematic framework that integrates key elements (Chang et al., 2005). There is a particular lack of in-depth analysis of the interactive mechanisms between reviewer identity characteristics (such as expertise and similarity) and individual consumer differences (such as risk preference and cultural background). Furthermore, the moderating role of platform environmental factors (such as community atmosphere and algorithmic presentation methods) on review perception has not been systematically elucidated. Furthermore, existing methodological frameworks require refinement: while questionnaire surveys efficiently gather large-sample data, they necessitate enhanced validation of scale validity and control for common method bias; experimental studies may reveal causal relationships but should improve scenario authenticity and external validity (Li & Liang, 2022). These theoretical gaps hinder a comprehensive understanding of the complex, holistic influence of online reviews on decision-making, thereby undermining the development of effective review management strategies and consumer information discernment practices.

Given the aforementioned gap, this study aims to systematically elucidate the multifaceted factors influencing consumer purchasing decisions through online reviews and their underlying mechanisms. The core research question (RQ) is: What key factors within online reviews, and how do these factors interact, collectively influence consumer purchasing decisions? To comprehensively and objectively synthesize the current highly fragmented research findings, this study employs a combined methodology of systematic review and bibliometric analysis. Throughout the research process, we shall uphold methodological neutrality, conducting systematic quality assessments of included studies' research designs. This approach avoids a priori biases toward any specific methodology (e.g., questionnaires, experiments), ensuring conclusions are grounded in objective analyses of methodological rigor and applicability. The systematic review facilitates the systematic identification, screening, and evaluation of existing empirical literature, enabling a qualitative synthesis of influencing factors and causal pathways. Bibliometric analysis, meanwhile, objectively maps the knowledge structure, evolutionary trajectory, and research hotspots within the field, identifying key authors, publications, and potential collaborative networks. This approach thus addresses the subjective limitations of traditional reviews, providing a scientific foundation for constructing a holistic theoretical framework.

The overall structure of this study is organized as follows: first, a comprehensive and systematic literature review is conducted on the subject under investigation, mapping existing research findings and their contextual development; subsequently, the materials employed, and the overall methodological framework adopted throughout the research process are elaborated in detail. As the study progresses, valuable insights and perspectives on the current state of knowledge development within this field are distilled, drawing upon the outcomes of bibliometric analysis, the conceptual framework constructed, and in-depth discussion and analysis. Finally, the study systematically summarizes the principal findings and proposes a research agenda for future investigation, grounded in the existing body of work.

To clearly illustrate the comparison between this study and relevant cutting-edge work, Table 1 provides a systematic comparison across three dimensions: theoretical framework construction, methodological innovation, and research foresight.

Although existing literature—such as the systematic review of consumer digital information sharing in Reference Lei et al. (2021) and the bibliometric analysis of digital marketing and consumer behavior in Reference Li et al. (2023)—has laid the groundwork for the field, this study distinguishes itself by focusing on the core element of 'online reviews' and constructing an integrated analytical framework for the 'decision-making ecosystem' encompassing seven theoretical dimensions. This framework transcends the traditional 'input-process-output' model employed in Reference Lei et al. (2021), revealing a multi-layered, dynamic mechanism of action spanning from foundational feature extraction to online trust construction, and further to psychological mechanisms and misinformation detection. Moreover, compared to the broad research agenda proposed in Literature 2—encompassing social media, AI, VR, and other diverse topics—this study offers a more refined, data-driven deconstruction of online review influence mechanisms. It explicitly proposes future research directions represented by 'large language model-driven intelligent analysis systems' and 'cross-cultural dynamic prediction models'. Thus, this study's core contribution lies in bridging the fragmentation of existing theoretical research, offering enterprises navigating the challenges of online reviews in the digital economy a new paradigm that combines theoretical depth with practical feasibility.

2. Methodology

This study employs a mixed methodology combining systematic literature review with bibliometrics and content analysis. This approach integrates the strengths of both quantitative and qualitative analysis, enabling it to reveal the field's knowledge structure and evolutionary trends at a macro level through scientific maps, while simultaneously elucidating its underlying mechanisms at a micro level through in-depth content analysis. This ensures the rigor, comprehensiveness, and depth of the research process (Mengist et al., 2020).

Table 1
Comparison between the latest and most advanced research.

Author	A systematic, multi-layered theoretical framework for the "decision-making ecosystem" has been constructed, effectively integrating existing fragmented research.	A clear, reproducible path for bibliometric analysis has been provided and validated, serving as a model for research methodologies within this field.	It reveals the critical balancing mechanism between data utilisation and privacy protection and charts a new research paradigm for the future driven by cutting-edge technologies such as large language models.	Main research activities
Cruz-Cárdenas et al. (2025)	×	×	×	This study systematically reviews and synthesizes consumer information and knowledge-sharing behaviors within digital contexts. Employing the PRISMA methodology, 51 studies were screened. Content analysis combined manual categorization (input-process-output model) with artificial intelligence tools, revealing a predominance of quantitative methods focused on individual consumer and psychological characteristics. Analysis identified five thematic clusters broadly consistent with the behavioral model adopted, subsequently proposing future research directions and practical recommendations. This study fills a gap in consumer behavior research by providing a systematic review of the digital I&KS domain.
Chu et al. (2025)	×	×	×	This paper explores how digital media is transforming the way service industries—particularly hospitality, tourism, and travel—interact with customers, thereby proposing a future research agenda. Focusing on six core domains—online reviews, social media, artificial intelligence, virtual/augmented reality, influencers, and service marketing theory—it identifies key research directions. Furthermore, it charts a roadmap for future service industry research across three dimensions: emerging technologies, cross-cultural studies, and ethics and diversity.
Figueiredo et al. (2025)	×	√	×	This study employs systematic bibliometric analysis to synthesize research on the impact of digital marketing on online consumer behavior into a six-theme framework. Theoretically, it constructs a structured cognitive system; methodologically, it ensures the verifiability of conclusions through objective analysis. Building upon this foundation, it proposes forward-looking research directions, offering significant guidance for the future development of this field.
Ours	√	√	√	This study employs bibliometric analysis to systematically trace the research trajectory of online reviews influencing consumer decision-making, constructing a multi-layered theoretical framework integrating seven dimensions. It extends from foundational analysis to trust-building and decision-making ecosystems. The research not only reveals balancing mechanisms between data technology and privacy protection but also proactively proposes developing large language model-driven intelligent analysis and cross-cultural predictive models, charting a course for innovative precision marketing research paradigms.

2.1. Data sources and retrieval strategy

To enhance the comprehensiveness and robustness of literature retrieval, this study draws its data from an internationally authoritative peer-reviewed literature database: the Web of Science (WoS) Core Collection. This database was selected for its extensive coverage of high-impact journals in the social sciences, its standardized data quality, and its complementary nature, collectively providing a more reliable and comprehensive data foundation for bibliometric analysis. The retrieval strategy was meticulously designed to optimize both recall and precision. Thematic retrieval employed an optimised Boolean logic query. To accommodate the retrieval rules of different databases, the query was executed within the Web of Science (WoS) based on the 'Topic' (TS) field. The core retrieval logic is as follows: (((‘online review*’) AND (“e-WOM” OR ‘electronic word-of-mouth’ OR ‘online comment*’ OR ‘user review*’)) AND ((‘online shopping *’) AND (‘purchase intent *’ OR ‘buying decision*’ OR ‘consumer decision*’ OR ‘online purchase*’ OR ‘buy* online’ OR ‘e-commerce’ OR ‘electronic commerce’)). The query employs the asterisk (*) as a truncation symbol to cover all variants of relevant terms (e.g., the plural form 'reviews' for

"review"), and broadly encompasses core synonyms for 'online review' (e.g., 'e-WOM'). Retrieval for each database was executed on the same date, 2 May 2025, to ensure temporal consistency. Publication dates were restricted to 1 January 2000 to 31 December 2024 to analyze the field's comprehensive development over the past two decades.

2.2. Inclusion and exclusion criteria

To identify the most relevant high-quality literature, we established clear criteria: Inclusion criteria: (1) Peer-reviewed English-language journal articles; (2) Research core focused on examining the impact of online reviews on consumer purchasing decisions; (3) Published between 1 January 2000 and 31 December 2024 (Note: The search date was 2 May 2025; hence the data year was restricted to complete years prior to the search date, i.e., 2024 and earlier, for greater rigor). Exclusion criteria: (1) non-research literature such as conference abstracts, book reviews, editorials, and news reports; (2) Literature demonstrably unrelated to the research topic (e.g., focusing on B2B decision-making or offline word-of-mouth); (3) Literature formally retracted.

2.3. Literature screening and data cleaning process

First, deduplication and merging: import the search results from a database into the literature management software Zotero and utilize its automated duplicate detection feature combined with manual verification to remove duplicate records. Subsequently, preliminary screening (title/abstract screening) was conducted: Three researchers independently reviewed the titles and abstracts of all remaining literature, performing an initial screening based on predefined inclusion/exclusion criteria. Inclusion criteria were: (1) peer-reviewed English-language journal articles; (2) research core focused on empirically or theoretically examining the impact of online reviews (or synonymous concepts) on consumers' (individual) purchasing decisions, intentions, or behaviors; (3) publication within the specified timeframe. Exclusion criteria were: (1) non-research literature such as conference abstracts, book reviews, editorials, or news reports; (2) literature where the research subjects were not end consumers (e.g., B2B decision-making) or the context did not involve online reviews (e.g., focusing on offline word-of-mouth or seller-generated content); (3) literature where the full text was unavailable or had been formally retracted. Disagreements among researchers were resolved through discussion or adjudication by a third researcher.

Subsequently, a full-text review was conducted: the Full texts of papers that passed the initial screening were obtained and independently scrutinized by the same three researchers to definitively determine their eligibility. Disagreements were resolved through discussion or third-party arbitration.

To objectively assess inter-rater reliability during literature screening, we conducted an Intraclass Correlation Coefficient (ICC) analysis on the initial screening outcomes of the three researchers. Calculations employed the ICC (3, *k*) metric, which is suitable for a fixed-rate model, yielding an ICC value of 0.9247. This value is very close to 1, indicating extremely high inter-rater reliability among the three assessors when evaluating literature relevance (Koo & Li, 2016). This methodologically enhances the rigor of the literature screening process and the credibility of its outcomes.

Finally, data cleansing and standardization: To ensure the accuracy of the bibliometric analysis, the metadata of the final included literature (such as author keywords, source publication names, and author names) underwent thorough cleansing and standardization. For instance, singular and plural forms (e.g., 'review' and 'reviews'), British and American spellings (e.g., "behavior" and 'behavior') were consolidated, abbreviations were standardized, and obvious data entry errors were corrected. Following this rigorous process, 256 articles were unanimously deemed compliant with the criteria and incorporated into the final literature repository, forming the foundation for subsequent analysis.

2.4. Analytical methods

To comprehensively address the research questions, this study employs a hybrid methodological framework combining bibliometric analysis with content analysis.

(1) Bibliometric Analysis

Two primary techniques were utilized: keyword co-occurrence analysis and co-citation analysis. VOSviewer (version 1.6.20) software was employed for network mapping and cluster analysis. This analysis aimed to reveal objectively, at a macro level, the field's research hotspots, knowledge foundations, and disciplinary structure (corresponding to the identification of 'key factors' within the research question).

(2) Content Analysis

To address the limitations of bibliometrics in deeply elucidating the complex interplay of factors, this study incorporated content analysis. Building upon the key clusters and core literature identified through bibliometric analysis, researchers conducted a close reading and qualitative analysis of representative, highly cited, and emerging frontier literature within each cluster. The analysis focused on extracting and synthesizing the intrinsic mechanisms, mediating variables, moderating variables, and theoretical explanations within the literature regarding how and why various factors of online reviews (such as sentiment polarity and reviewer expertise) influence consumer decision-making. This analysis aims to provide an in-depth interpretation and synthesis of the bibliometric findings. It seeks to offer a micro-level qualitative explanation of the mechanisms through which these factors interact and jointly influence outcomes (directly addressing Research Question RQ2). Furthermore, it engages with existing theories to construct an integrated theoretical framework.

2.5. Bibliometric analysis process

This study employs bibliometric methodologies, utilizing VOSviewer 1.6.20 software to comprehensively apply keyword co-occurrence analysis and co-citation analysis. This approach systematically reveals thematic connections, knowledge structures, and research frameworks within the domain of 'online reviews influencing consumer purchasing decisions'. The combination of these two approaches stems from their inherent complementarity: keyword co-occurrence analysis illuminates the frontier themes and conceptual connections within current research, while co-citation analysis traces the field's knowledge foundations and theoretical lineage. Together, they form a multidimensional analytical perspective encompassing 'horizontal conceptual networks and vertical knowledge transmission'.

The minimum occurrence frequency for keywords and cited literature was set at five instances. This threshold was established based on conventional practices in bibliometrics and the data characteristics of this study: it ensures the analyzed subjects possess sufficient academic influence or representativeness while avoiding the loss of important information due to an excessively high threshold or the introduction of noise interference from an excessively low threshold. Specifically, frequency distribution analysis revealed that a threshold of five occurrences effectively covers the core literature set (comprising approximately 15–20% of the total), preserving key knowledge nodes while maintaining network structural clarity.

Using the association strength algorithm, six keyword clusters and seven co-citation clusters were generated, respectively. The number of clusters was not predetermined but based on the automatic optimization of network modularity achieved by the VOSviewer clustering algorithm. The variation in the number of documents within each cluster reflects

the natural distribution characteristics of the knowledge structure within this field, rather than being determined by an artificially set fixed threshold. The specific parameter settings are as follows:

Keyword co-occurrence analysis aims to identify core themes and hot topics within this research field. During analysis, 'keywords' serve as the unit of analysis, with a minimum occurrence threshold of five instances set. Co-occurrence strength is calculated using the association strength algorithm. The network is automatically partitioned via VOSviewer's built-in clustering algorithm, ultimately generating a visualized map comprising six clusters. This cluster count is automatically determined by the software according to the principle of maximizing network modularity, ensuring the scientific rigor and objectivity of the classification.

Co-citation analysis of literature aims to explore the solid knowledge base and classic literature groups within a field. In the analysis, 'cited references' serve as the unit of analysis, with a minimum citation frequency of five set for each document. By calculating the co-citation strength between documents, a network of connections is constructed, forming a knowledge structure map comprising seven major clusters. The number of clusters is likewise determined based on the algorithm's automatic identification of the network structure.

This methodological combination offers distinct advantages: on the one hand, diachronic analysis of co-cited literature enables the identification of generational succession within academic networks, capturing implicit knowledge connections between research communities (Candeias Fernandes & Franco, 2022); on the other hand, synchronic analysis of keyword co-occurrence facilitates the construction of conceptual topological maps within research fields, revealing the distribution and intersecting trends of core themes (Zeng & Yang, 2023). The mutual corroboration of these two approaches not only reconstructs theoretical development trajectories but also visualizes the migration patterns of academic hotspots through knowledge visualization, thereby forming a multidimensional analytical model with profound explanatory power (Xia et al., 2023).

By clearly defining the aforementioned analytical tools, software versions, key parameters, and their configuration rationale, this study ensures transparency and soundness in the analytical process alongside reproducibility of results, thereby establishing a robust foundation for subsequent in-depth discussions (Fig. 1).

3. Results

3.1. Descriptive analysis

As shown in Fig. 2, the number of research papers on the impact of online reviews on consumer purchasing decisions was at a relatively low

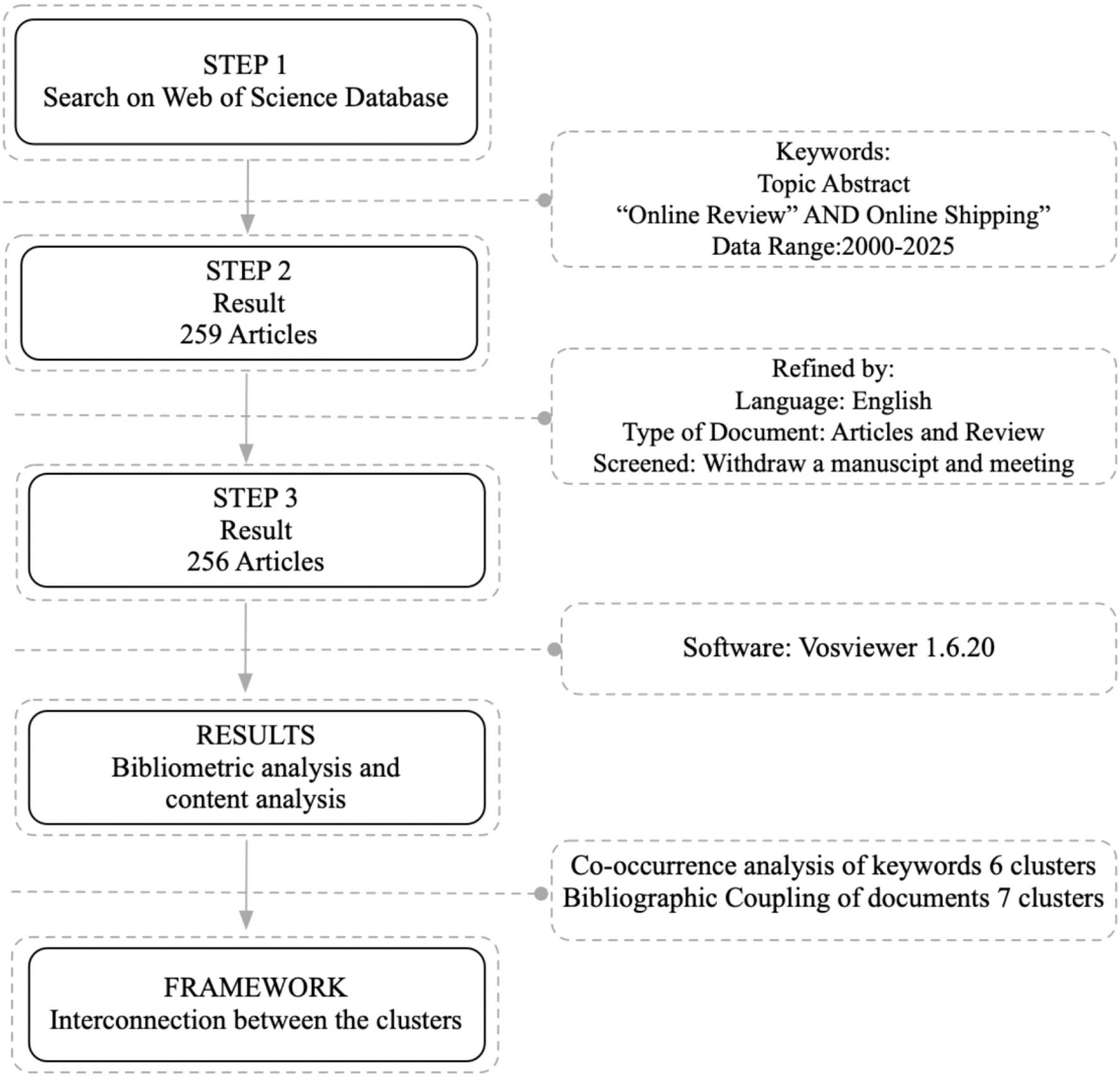


Fig. 1. Research program.

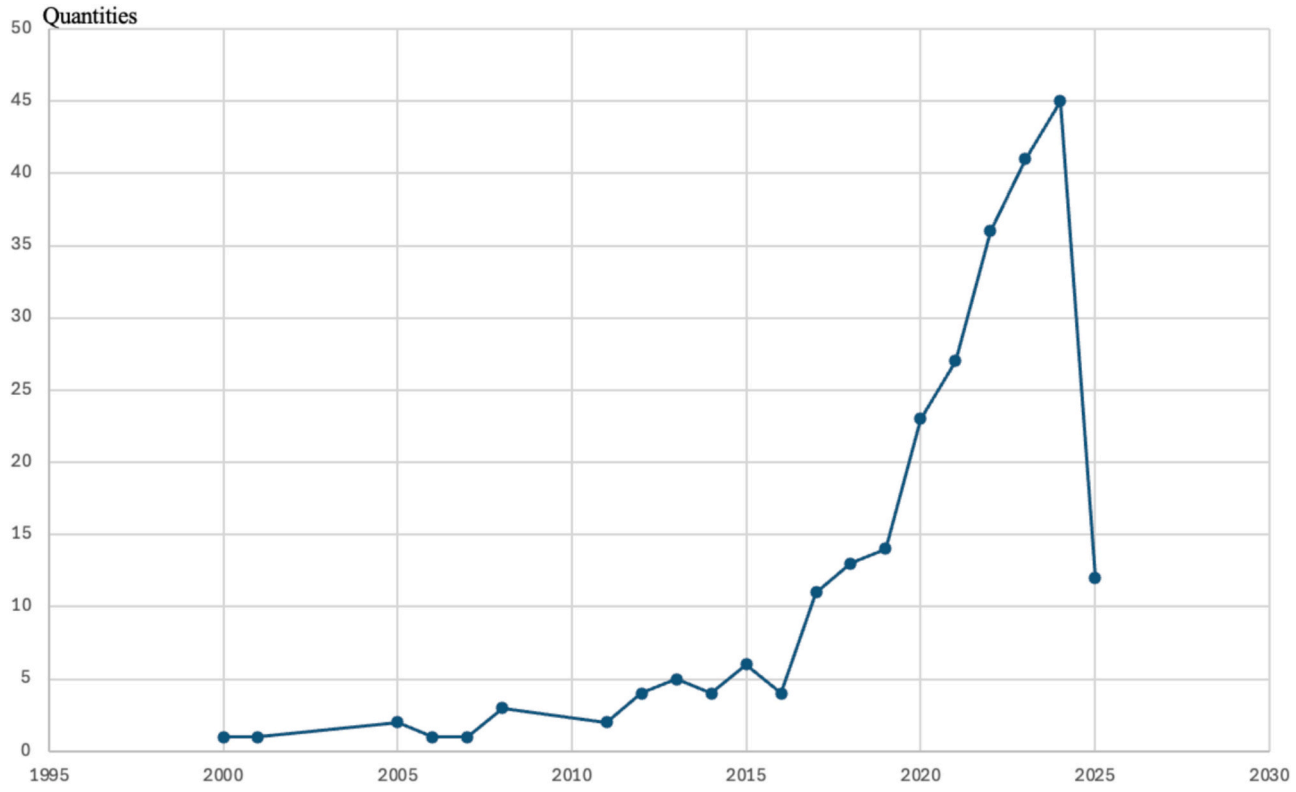


Fig. 2. Plot of the number of papers vs. time.

level overall and grew slowly between 2000 and 2015, with some minor fluctuations during this period. This may indicate that research in this area was conducted during this period, when online shopping had not yet become a mainstream mode of consumption, which limited the actual scenarios of the impact of online reviews on consumer purchases. However, after 2015, as e-commerce platforms became more sophisticated, the number of relevant papers published gradually increased, suggesting a growing interest in the topic among researchers. In particular, the number of publications reaches a significant peak after 2024, which may reflect the fact that the study of the impact of online reviews on consumer behavior has attracted a lot of attention to this area of research in this year, which may be related to the rapid development of e-commerce, the deepening of the study of consumer behavior, as well as the popularity of online review platforms.

The distribution of articles by country is illustrated in Fig. 3. The figure clearly demonstrates that China's contribution significantly surpasses other nations, reaching nearly 140 articles and forming a prominent peak. The United States ranks second with approximately 40 articles. Contributions from other countries are relatively limited, with most below 20 articles and a few nations contributing only single-digit figures. This preliminary distribution indicates that China and the United States are the primary research forces in this field, exhibiting high academic activity.

However, this distribution may be influenced by publication bias and

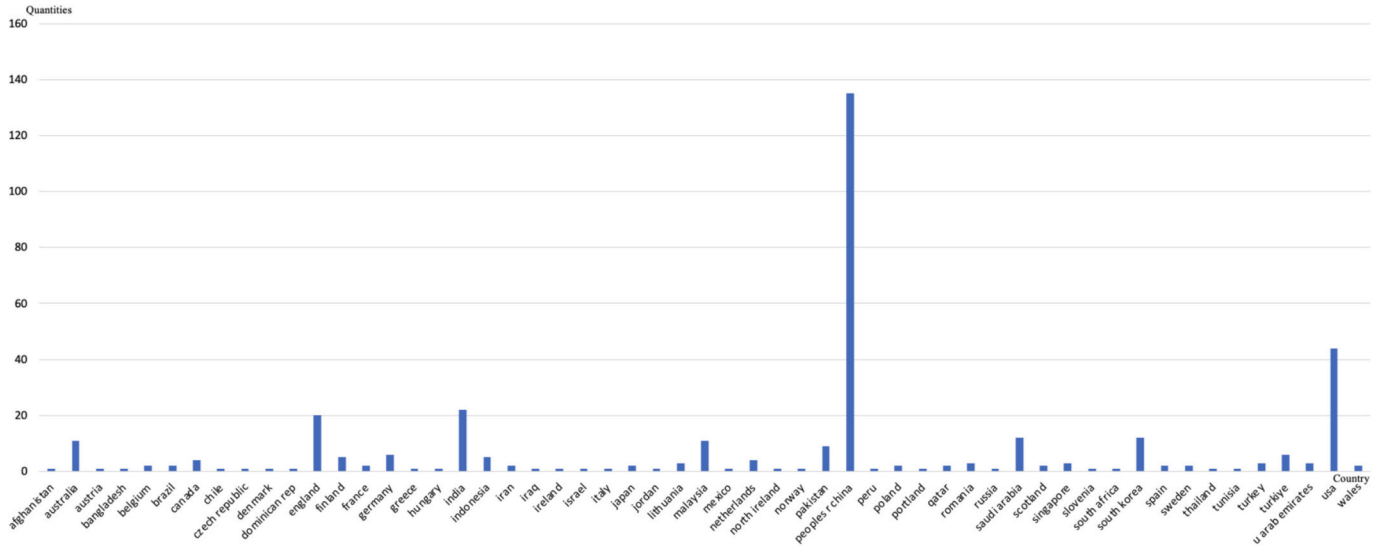


Fig. 3. Number of articles by country.

limitations in the language of literature retrieval. Firstly, the national distribution depicted may reflect imbalances within the academic publishing ecosystem. For instance, international journals primarily publishing in English are more readily indexed by databases, potentially excluding high-quality research from non-English-speaking countries due to language barriers, thereby underestimating their actual contributions. Secondly, the databases relied upon in this analysis predominantly feature English-language literature, failing to systematically incorporate significant contributions in other languages such as Chinese and Spanish. This may result in underrepresentation of representative research from non-English-speaking nations like China, thereby affecting the comprehensiveness of comparative assessments of research output across countries. Consequently, when interpreting variations in national distribution, it is essential to recognize that these reflect not only actual research activity but may also be subject to systematic influences such as literature accessibility, publication preferences, and linguistic coverage.

According to the distribution chart of the number of journals and papers published, as shown in Fig. 4, the distribution of papers on the impact of online reviews on consumer purchase decisions is demonstrated. By analyzing the chart, we can observe the concentration and distribution characteristics of papers published. Among them, Sustainability published 16 articles showing the journal's high concern and support for the topic of online reviews affecting consumers' purchase intention, followed closely by Electronic Commerce Research And Applications with 14 articles, IEEE Access and internet research with 12 articles, indicating that these journals have also shown greater interest in this research topic, as well as Applied Sciences-Basel and Decision Support Systems with 7 articles, Heliyon and Information Processing & Management with 6 articles, and other journals with 6 articles. Management published six articles, while the rest of the journals published fewer than five articles, indicating that there is still room for expansion in this area of research across a wide range of journals.

As illustrated in Fig. 5, the evolution of academic influence within this field exhibits pronounced fluctuations, vividly reflecting pivotal junctures in its developmental trajectory. Quantitatively, the total number of paper citations reached its first peak in 2010 (nearly 1000 citations), signifying the formation and widespread recognition of foundational theories within the domain. Subsequently, citations plummeted to a trough in 2015 (approximately 200 citations), suggesting the research entered a period of assimilation and accumulation. The second peak observed in 2020 (nearly 800 citations) is highly indicative. Qualitatively, this represents not mere repetition, but a paradigm shift driven by dual forces: the empowerment of new-generation technologies (such as deep learning applications in sentiment analysis) and global societal transformations (such as the COVID-19 pandemic, which has propelled online consumption to mainstream status). This convergence catalyzed a wave of new, highly influential research outputs. The overall trend indicates that this field is a dynamic discipline highly responsive to external technological and societal factors, with its influence peaks closely aligned with generational shifts in theory or technology.

The five-year impact factor distribution of journals contributing to this study's literature sample exhibits a broad spectrum and pronounced hierarchical structure. This quantitative characteristic qualitatively reveals the interdisciplinary nature and developmental trajectory of this research field, as illustrated in Fig. 6. Specifically, impact factor values span a broad range from single digits to nearly 30, with the vast majority of journals concentrated below 10. Top-tier review journals such as ACM Computing Surveys (five-year IF ~27) occupy the apex of this influence distribution. This pyramid-like structure indicates that the field of 'Online Reviews and Consumer Decision-Making' not only possesses an authoritative theoretical foundation with long-term influence, recognized by the academic community (represented by high-tier journal publications), but also constitutes a vibrant, extensive knowledge system comprised of numerous studies published in core journals across

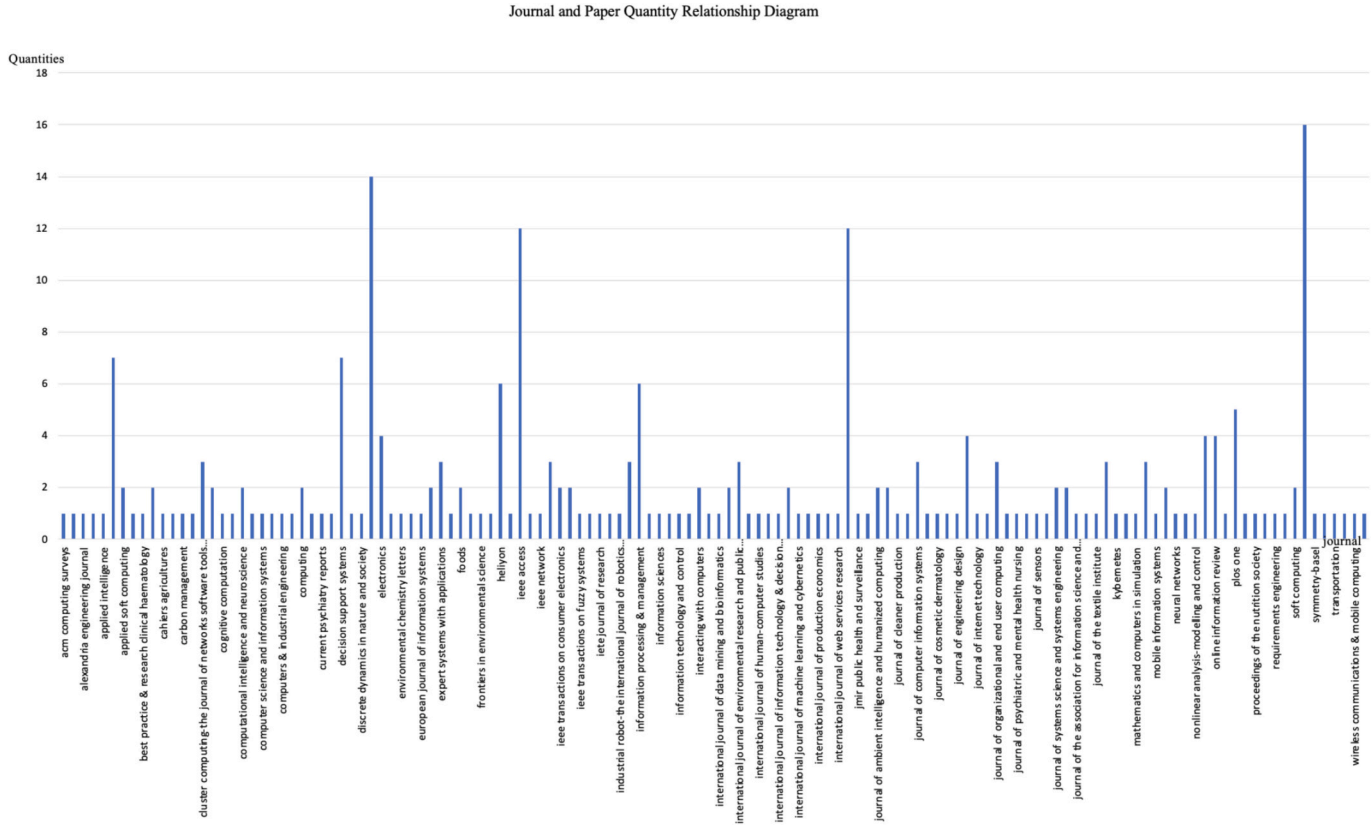


Fig. 4. Relationship between the number of journals and papers.

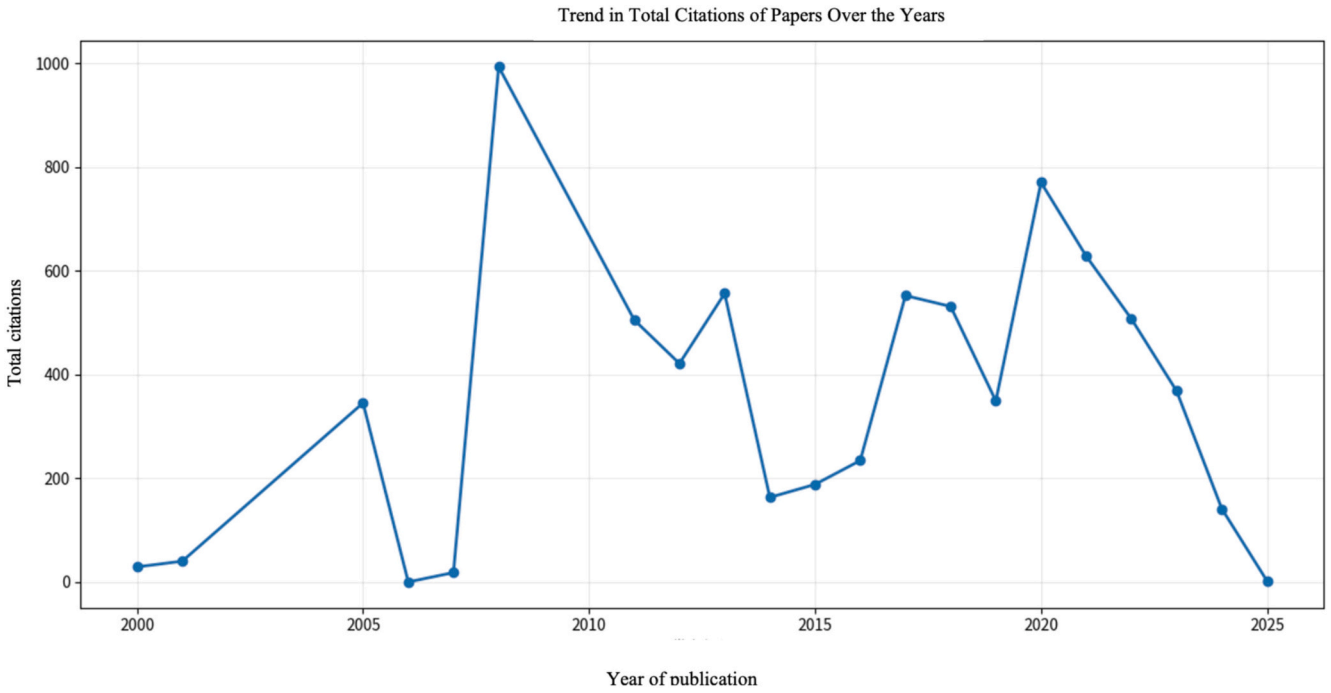


Fig. 5. Trend chart of total citations for papers over the years.

various specialized fields. The broad spectrum of journal impact reflects the field's high degree of interdisciplinarity, attracting contributions from scholars across multiple disciplines ranging from computer science and information management to marketing and consumer behavior studies. Consequently, the sample repository used in this review is not a random collection but a representative sample that combines depth (pioneering works from high-impact journals) with breadth (specific findings from a wide range of journals). This ensures subsequent analyses comprehensively and reliably reflect the knowledge structure of this field.

3.2. Descriptive analysis

This study uses two bibliometric analysis techniques, keyword co-occurrence and bibliographic coupling, to construct a comprehensive and integrated analysis perspective for the research field from multiple dimensions. To ensure the consistency and professionalism of the analysis, all bibliometric analyses were conducted using VOSviewer software version 1.6.20 (Bukar et al., 2023).

3.2.1. Keyword co-occurrence analysis results and discussion

Keyword co-occurrence analysis effectively reveals the core thematic structure within the domain of "online reviews influencing consumer purchasing decisions." As shown in the network diagram (Fig. 7), node size represents the frequency of keywords, line thickness indicates the strength of co-occurrence, and distinct colors clearly delineate the six primary research clusters that VOSviewer automatically clustered.

First, examining the network's overall structure, nodes such as "online shopping" (40 occurrences) and "word of mouth" (36 occurrences) occupy relatively central positions in the graph and are densely connected to other nodes. This indicates that "online shopping behavior" and "digital word of mouth" represent long-standing, stable core topics in this field, forming the foundational focus of researchers' collective attention. A notable trend is the prominent scale of the red cluster, centered around keywords such as "sentiment analysis" (33 mentions), "machine learning," and "deep learning." This reflects how artificial intelligence technologies—particularly mature natural language processing techniques—are powerfully driving review analysis research away from traditional content analysis toward intelligent, automated approaches, establishing it as the most dynamic research frontier.

Secondly, the network diagram clearly illustrates the intersections and evolution of research themes. For instance, the "Covid-19" node forms tight connections with nodes like "online shopping" and "behavior" (yellow cluster), directly demonstrating how sudden public health crises, as powerful external drivers, accelerated the shift of consumer behavior online and spurred concentrated discussions on the psychological and behavioral mechanisms of online consumption during exceptional periods. Additionally, the green cluster formed by "e-commerce" (25 occurrences), "social commerce," and "trust" highlights that, beyond traditional e-commerce models, trust-building in "social commerce" based on social relationships has become a significant emerging branch. This shift is directly driven by the vigorous development of new industry formats, such as live-streaming sales and community marketing.

In summary, this co-occurrence network not only statically displays the conceptual architecture of the field but also dynamically reveals its evolutionary dynamics: on one hand, paradigm shifts driven by technological innovations (such as AI) at the methodological level; on the other, new challenges posed by industry practices (such as social commerce and pandemic impacts) at the practical level. These six clusters collectively outline a dynamic research landscape that remains grounded in classical consumer behavior theory while maintaining high responsiveness to external technological, social, and industry shifts.

In this co-occurrence analysis of keywords, all high-frequency keywords were grouped into six clusters with distinct semantic boundaries (Table 2). In terms of scale distribution, Cluster 1 (Sentiment Analysis, 29%), Cluster 2 (E-commerce, 21.5%), and Cluster 3 (Influence Mechanisms, 20.4%) collectively form the core knowledge system of this field, accounting for over 70% of the total. This reflects the three pillars of research: technological orientation, platform-based delivery, and effectiveness evaluation. Cluster 4 (Online Shopping, 13.9%) focuses on specific consumption scenarios. In contrast, Clusters 5 (Word-of-Mouth Communication, 7.5%) and 6 (Products and Services, 7.5%) represent peripheral yet specialized research directions, indicating a trend toward deepening development within the field. These six clusters clearly delineate the complete value chain of the "online reviews-consumer

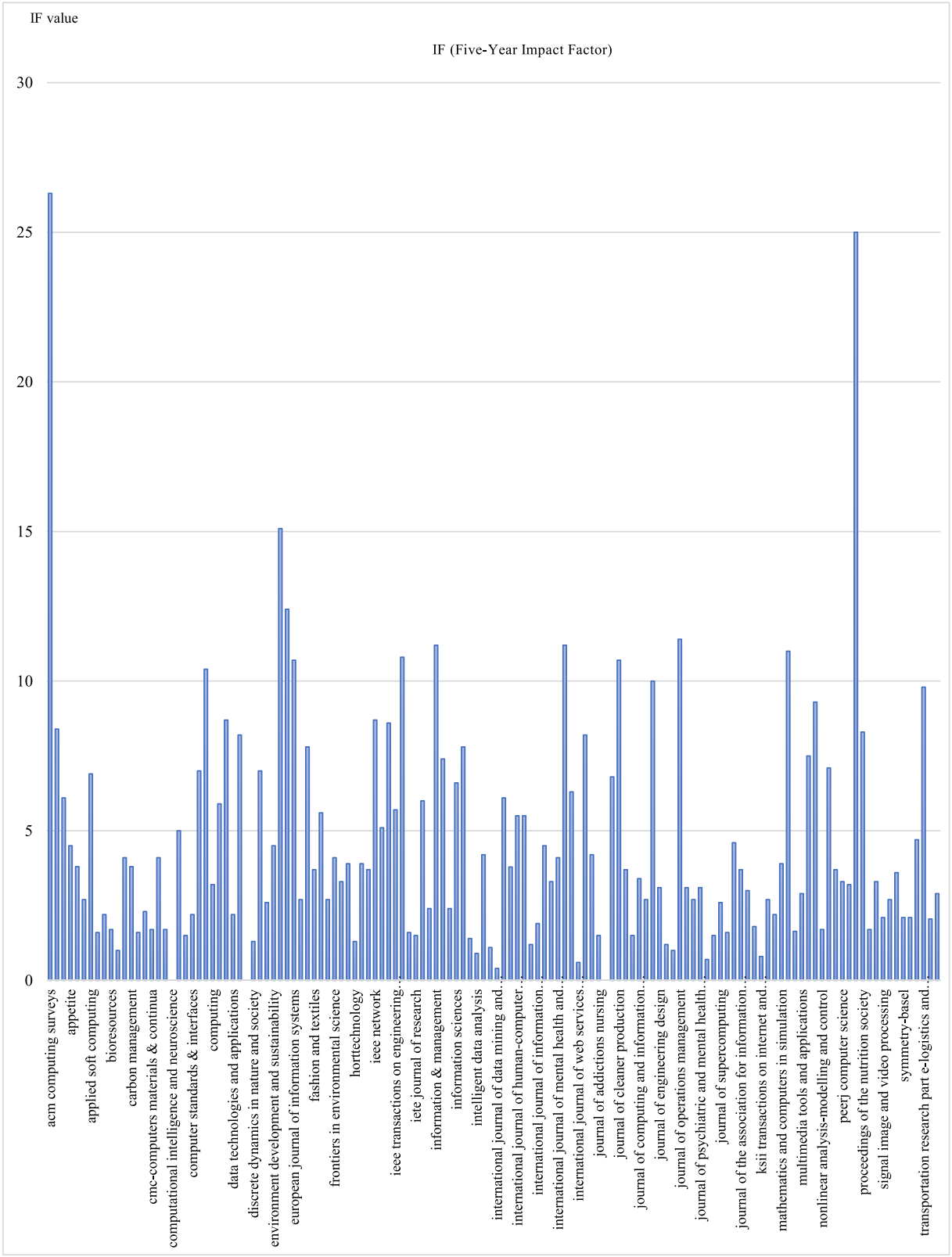


Fig. 6. Five-year impact shadow plot for journals.

decision-making" domain, ranging from technological enablement (Cluster 1) to psychological mechanisms (Clusters 2 and 3), and then to scenario applications (Cluster 4) and business management (Clusters 5 and 6). This structure illustrates how academic research responds to technological innovation, social change, and industry practices in a multidimensional manner.

To this point, this study has preliminarily addressed part of the core research question (RQ1): Key factors in online reviews can be identified across multiple dimensions, including the review content itself (sentiment, quality), the communication medium (platform, social

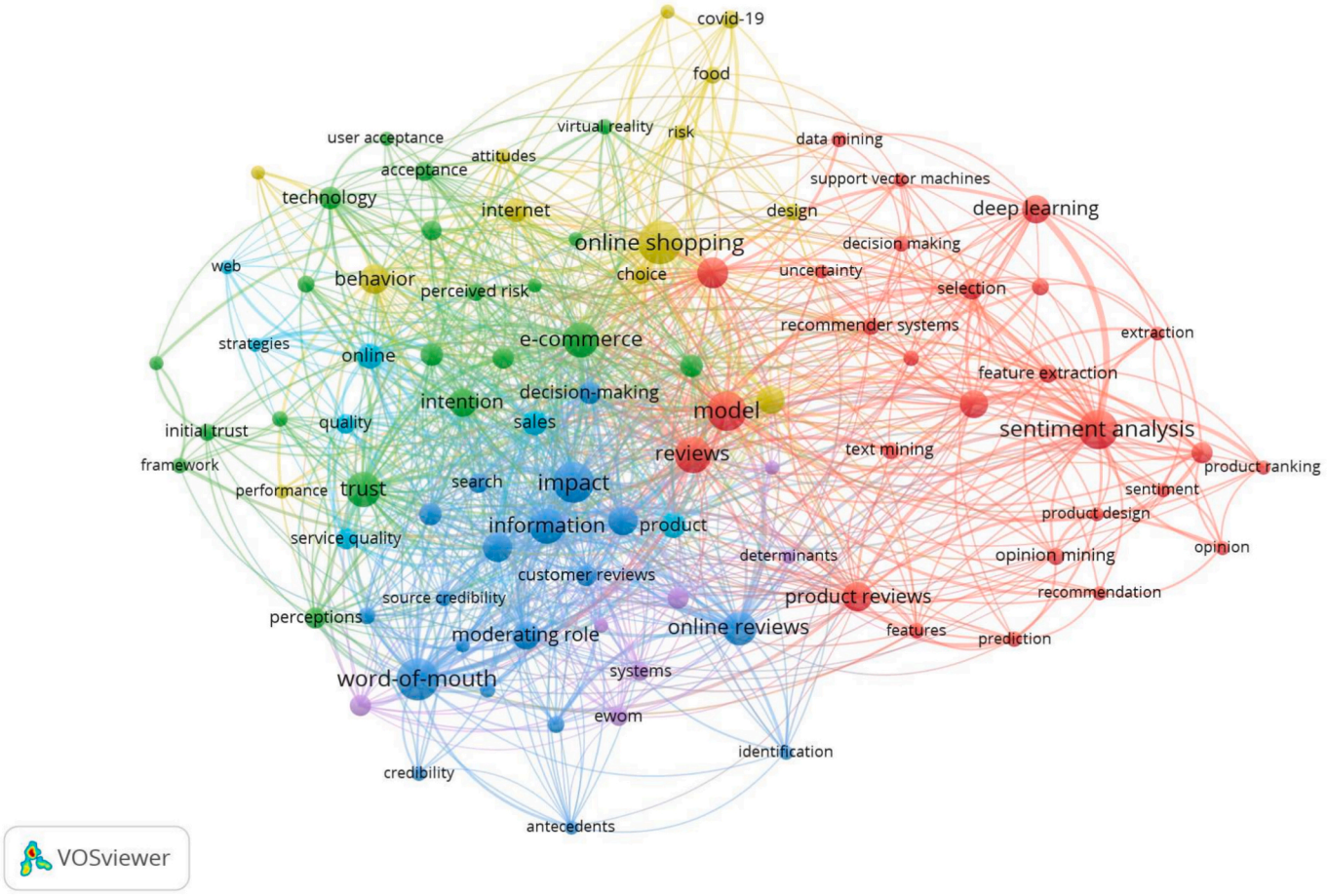


Fig. 7. Co-occurrence-based keyword network.

relationships), and the consumption context. These themes lay the groundwork for subsequent in-depth analysis of the interactive mechanisms among these factors.

3.2.2. Co-citation analysis of literature: knowledge base and theoretical evolution

This study employs co-citation analysis to explore the knowledge base and development trends in the field of online reviews influencing consumer purchasing decisions. Using 256 obtained documents as the analytical foundation, a co-citation network was constructed via VOSviewer software. The analysis treated "cited references" as the unit of analysis, with each cluster requiring at least seven documents to ensure result concentration and validity. Ultimately, the literature was divided into seven distinct clusters, with their knowledge structures depicted in Fig. 8. Within this co-citation network map, node size corresponds to citation frequency, while line thickness reflects the strength of co-citations between documents. The overall map exhibits a polycentric, networked structural characteristic. The seven clusters are clearly marked with different colors. For example, key literature nodes such as O'Brien and Toms (2008), Rotem Mindali and Weltevreden (2013), and Gao et al. (2012) demonstrate the field's diverse research themes and intersecting knowledge foundations. This provides an intuitive basis for understanding its theoretical origins and evolutionary trajectory.

A detailed analysis reveals that the network exhibits a distinct "core-periphery" structure and diachronic distribution characteristics. Literature situated in the central region of the map (e.g., O'Brien and Toms (2008) on user engagement, Gao et al. (2012) on information processing in online decision-making) features larger nodes and denser connections, indicating their role as foundational theories linking diverse research directions with broad and enduring influence on the field. A notable trend is the emergence of purple and blue clusters, as represented by Yang, Ouyang, et al. (2020) and Coffino et al. (2020), although positioned relatively peripherally on the map, are characterized by recent publication dates (around 2020) and strong interconnections. This reflects how recent applications of AI technology in sentiment analysis (e.g., deep learning-based review mining) and the impact of sudden public events (e.g., the COVID-19 pandemic) on consumer behavior have emerged as new focal points driving the discipline's frontier development.

Furthermore, the spatial relationships between clusters reveal the directions of knowledge flow. For instance, early studies focused on theoretical frameworks (e.g., Chang's et al. (2005) model on online shopping adoption) cluster on one side of the map, while research on platform trust and social media (e.g., Lee et al., 2011, Zhang et al., 2018) forms a distinct knowledge cluster on the opposite side. Studies on user behavior and psychological mechanisms (e.g., O'Brien & Toms, 2008) bridge these two clusters. This arrangement intuitively illustrates the field's knowledge evolution trajectory—from "technology adoption" to "behavioral psychology" and finally to "scenario applications." This progression stems from the industry shift where e-commerce models transitioned from simple transaction platforms to social, immersive experience platforms.

In summary, the co-citation network not only statically identifies seven knowledge clusters but also dynamically reveals the diffusion pathways of the knowledge base from classical theories to peripheral frontiers, reflecting the proactive response of academic research to technological innovation and real-world societal transformations.

Through an in-depth interpretation of the essence of clustering, this

Table 2
Clusters of high-frequency keywords in online review research.

Cluster number	Color	Cluster name	Main keywords
1	Red	Sentiment analysis	Algorithm, classification, data mining decision making, deep learning, electronic commerce extraction, feature extraction features, machine learning, model, natural language process, opinion, opinion mining, prediction, product design, product ranking, product reviews, recommendation, recommender systems, selection, sentiment, sentiment analysis, support vector machine
2	Green	E-commerce	Acceptance, adoption, attitude, consumer behavior, customer satisfaction, e-commerce
			Experience, framework, initial trust, intention perceived risk, perceptions, purchase, social commerce, systematic literature, technology trust, virtual reality
3	Blue	Impact	Antecedents, consumer reviews, credibility, consumer reviews, decision-making, identification, impact, Information, moderating role, online consumer review, perceived helpfulness, purchase intention responses, satisfaction, search, source credibility, word-of-mouth
4	Yellow	Online shopping	Attitudes, behavior, choice, covid-19, design, food, internet, online shopping, performance, risk, social media, support, systematic review
5	Purple	Word of mouth	Consumers, determinants, e-WOM, management, reputation, sustainability, systems
6	Cyan	Product	Online, product, quality, sales, service quality, strategies, web,

study systematically addresses the core research questions:

(RQ1) Primary Influencing Factors: Analysis identifies seven major categories of key factors affecting consumer decision-making, encompassing: Review Content Characteristics (emotional polarity, information quality, corresponding to Cluster 1) Source and trust characteristics (reviewer expertise, platform credibility, corresponding to Cluster 2), Social interaction and word-of-mouth (corresponding to Clusters 3 and 5), Consumer cognition and psychology (decision mechanisms, psychological interventions, corresponding to Clusters 6 and 7), Personalization and behavioral modeling (corresponding to Cluster 4), and macro-level product/service and sustainability factors (corresponding to Cluster 6).

(RQ2) Factor Interactions and Influence Mechanisms: Cross-connections between clusters in the graph indicate these factors synergistically influence decision-making through complex dynamic pathways. For instance, the technical outcomes of sentiment analysis (Cluster 1) form the foundation for understanding the characteristics of review content. These characteristics then impact trust-building (Cluster 2) and social interaction processes (Cluster 3), ultimately influencing consumer cognition and decision-making psychology (Clusters 6, 7). False review governance (Cluster 5) serves as a prerequisite for ensuring the effectiveness of all subsequent information processing stages (i.e., the data governance layer), while behavioral modeling and personalization (Cluster 4) demonstrate how platforms leverage these factors to optimize user experience. This indicates that influencing factors are not isolated but embedded within a dynamic system progressing from "data input and purification" to "information processing and trust establishment," then to "social influence and psychological decision-making," ultimately forming "behavioral output."

The seven clusters identified through analysis exhibit distinct and significant characteristics across multiple key dimensions: scale (measured by the number of articles within each cluster), influence (assessed by the total citation frequency of articles within each cluster), and research focus orientation. This strongly corroborates the breadth and diversity of topics encompassed within the field of investigating the impact of online reviews on consumer behavior in e-commerce. The following sections will elaborate on these clusters: (1) E-commerce application optimization based on sentiment analysis and product review mining (2) Technology-empowered and user-generated content-driven online trust building and cross-contextual consumption decision-making (3) Research on e-commerce loyalty generation mechanisms driven by multi-dimensional word-of-mouth interaction and trust synergy (4) Data-driven consumer behavior modeling and collaborative optimization of e-commerce strategies (5) Research on false review identification and review ecosystem governance driven by multidimensional data (6) Research on online consumption decision-making mechanisms driven by multidimensional information interaction (7) Research on psychological mechanisms and intervention strategies for online consumption behavior driven by user participation.

Cluster 1 (Sentiment Analysis and Intelligent Decision-Making), as

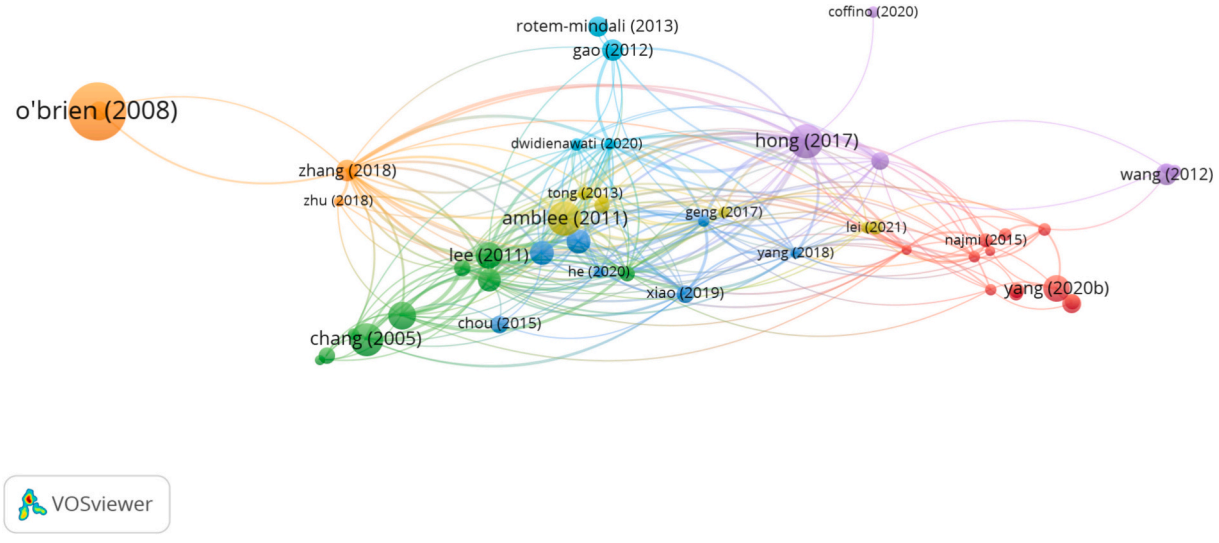


Fig. 8. Co-citation analysis of literature.

the most extensive and dynamic research direction in this field, reflects the core demands of the e-commerce industry's intelligent transformation. Progress in this domain is primarily driven by three major trends, which have yielded several representative technological achievements. The fundamental drivers are data surges and efficiency requirements, as shown in Table 3. Faced with massive volumes of online reviews, e-commerce platforms and brands urgently need automated, scalable review insights. Related research focuses on enhancing analytical efficiency and accuracy. For instance, the SLCABG model, which integrates sentiment lexicon and deep learning, significantly improves analysis performance in Chinese contexts (Yang, Li, et al., 2020). Meanwhile, the machine learning-based LSIBA-ENN model achieves refined sentiment classification (e.g., three-class) through innovative term weighting and feature selection methods, demonstrating outstanding classification performance and recall rates (Zhao et al., 2021). Extending the value chain from "understanding" to "application" represents a pronounced trend. Sentiment analysis has evolved beyond basic emotion detection to directly support high-value business decisions. For instance, extracting affective knowledge through NLP techniques and correlating product features with user perceptions provides methodological support for emotional product design (Jiao & Qu, 2019). Meanwhile, comprehensive product ranking systems effectively optimize ranking outcomes and online shopping experiences by synthesizing review information, thereby enhancing the quality of platform decisions (Najmi et al., 2015). The technological evolution addressing complex scenarios demonstrates strong problem-oriented characteristics. Research actively tackles specific challenges in real-world applications. For instance, to address the difficulty of comparing sentiment tendencies across platforms, a decision model integrating deep learning with the Q-ROFIWHM operator can effectively consolidate heterogeneous information and transform it into a reliable basis for decision-making (Yang, Ouyang, et al., 2020). In summary, the research within Cluster 1 represents not isolated technological iterations, but a direct response to core challenges encountered by the e-commerce industry during its digitalization and intelligent transformation—such as efficiency bottlenecks, decision support demands, and the need for complex scenario adaptation. Its technological development path remains highly synchronized with industry practice needs, forming a complete closed loop from methodological innovation to value creation.

Table 3
The top five authors of cluster 1.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
Sentiment analysis for e-commerce product reviews in Chinese based on sentiment lexicon and deep learning	Yang, Ouyang, et al. (2020)	IEEE Access	200	Quantitative analysis
A machine learning-based sentiment analysis of online product reviews with a novel term weighting and feature selection approach	Zhao and Li (2021)	Information Processing & Management	100	Quantitative analysis
A proposal for kansei knowledge extraction method based on natural language processing technology and online product reviews	Jiao and Qu (2019)	Computers In Industry	68	Quantitative analysis
A decision-making algorithm for online shopping using deep-learning-based opinion pairs mining and q-rung orthopair fuzzy interaction heronian mean operators	Yang, Li, et al. (2020)	International Journal of Intelligent Systems	66	Quantitative analysis
A comprehensive approach to product ranking using customer reviews	Karthik and Ganapathy (2021)	Computing	59	Quantitative analysis

To gain insights into the dynamics of customer online purchasing decisions, multiple studies have focused on key influencing factors, revealing distinct temporal trends and methodological evolution, as shown in Table 4. From 2005 to 2018, research emphasis shifted from foundational theory building to empirical testing of specific drivers, reflecting the e-commerce industry's progression from initial adoption to mature development. This trajectory aligns closely with industry drivers, including rising global internet penetration, mobile shopping growth, and expanding social media influence, particularly highlighting the central role of online reviews and trust mechanisms in consumer decision-making. Specifically, early studies employed qualitative analysis to construct reference models, laying theoretical foundations for understanding consumer behavior—their high citation counts underscoring their foundational value (Chang et al., 2005).

As the industry entered a period of rapid growth, empirical studies proliferated: Research on influencer recommendation scenarios revealed that the usefulness and trustworthiness of e-WOM significantly influence online purchasing decisions, moderated by variations in influencer reputation. This highlights the pivotal role of social media marketing in boosting conversion rates and reflects the industry's reliance on the influencer economy (Hsu et al., 2013). Concurrently, research on the integration of online reviews and advertising suggests that consumer purchase intent is influenced by the trustworthiness of the platform. In high-trust environments, reviews exert a stronger driving force—a finding aligned with e-commerce platforms' industry strategy of enhancing stickiness through trust-building (Lee et al., 2011). Another study confirms that online reviews surpass traditional reputation metrics in influencing store credibility, highlighting the dominance of user-generated content in consumer decisions—consistent with the industry's growing emphasis on word-of-mouth marketing (Utz et al., 2012). Regarding trust-building mechanisms, recent research identifies the synergistic effect of institutional trust and competence trust, moderated by shopping experience. This provides quantitative guidance for emerging businesses to optimize resource allocation, reflecting the industry's need for refined trust infrastructure operations (Stouthuysen et al., 2018). Collectively, these studies reveal an evolutionary trajectory from macro-level theories to micro-level mechanisms, with citation disparities reflecting greater practical influence of empirical research. Industry drivers—including intensified e-commerce platform competition, rising consumer demand for transparent information, and technology-enabled data-driven analysis—collectively propel research toward greater quantification and applicability. Future studies may deepen insights by incorporating regional variations (e.g.,

Table 4
The top five authors of cluster 2.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
Literature derived reference models for the adoption of online shopping	Chang et al. (2005)	Information & Management	305	Qualitative analysis
The effects of blogger recommendations on customers' online shopping intentions	Hsu et al. (2013)	Internet Research	211	Quantitative analysis
The different effects of online consumer reviews on consumers' purchase intentions depending on trust in online shopping malls an advertising perspective	Lee et al. (2011)	Internet Research	188	Quantitative analysis
Consumers rule: how consumer reviews influence perceived trustworthiness of online stores	Utz et al. (2012)	Electronic Commerce Research and Applications	146	Quantitative analysis
Initial trust and intentions to buy: the effect of vendor-specific guarantees, customer reviews and the role of online shopping experience	Stouthuysen et al. (2018)	Electronic Commerce Research and Applications	75	Quantitative analysis

comparisons between emerging and developed markets).

Research in this field, surrounding consumer digital behavior and decision-making, shows a clear trend of evolving from foundational loyalty building toward refined, scenario-based operations, as illustrated in Table 5. The research timeframe (2013–2019) coincided with the explosive growth of mobile e-commerce, the deep integration of social media, and the rise of cross-border consumption. This industry context shifted the research focus from general online loyalty to specific challenges, such as negative word-of-mouth management, cross-border consumption, female market segmentation, and incentive mechanism design. Specifically, early studies grounded in motivational theory revealed the underlying mechanism by which e-WOM participation influences online loyalty through platform identity formation, providing early theoretical support for e-commerce platforms to enhance user stickiness by building community identity (Yoo et al., 2013). As social media became a primary channel for expressing consumer sentiment, research shifted its focus to the generation mechanisms of negative word-of-mouth. Findings identified social psychological factors—such as face concerns and relationship strength—as key drivers, directly addressing businesses' urgent need for proactive online reputation risk management (Balaji et al., 2016).

Table 5
The top five authors of cluster 3.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
Determinants of negative word-of-mouth communication using social networking sites	Balaji et al. (2016)	Information & Management	160	Quantitative analysis
Exploring the effect of e-WOM participation on e-loyalty in e-commerce	Yoo et al. (2013)	Decision Support Systems	160	Quantitative analysis
The effects of online shopping context cues on consumers' purchase intention for cross-border e-commerce sustainability	Xiao et al. (2019)	Sustainability	92	Quantitative analysis
Female online shoppers examining the mediating roles of e-satisfaction and e-trust on e-loyalty development	Chou et al. (2015)	Internet Research	86	Quantitative analysis
Rebate strategy to stimulate online customer reviews	Yang and Dong (2018)	International Journal of Production Economics	38	Quantitative analysis

Research further deepened into specific consumption scenarios. Targeting the high-potential cross-border e-commerce market, studies using the Cue-Value Model have confirmed that situational cues indirectly drive purchase intention through both functional and emotional value, although brand familiarity exhibits negative moderation. This provides precise strategies for platforms to optimize marketing messages and overcome consumer decision inertia when entering emerging markets (Xiao et al., 2019). Another study precisely targets female consumers—a group characterized by high purchase frequency and significant decision influence. Using structural equation modeling, it demonstrates that privacy security, logistics timeliness, and interface design each drive e-loyalty through dual pathways of e-trust and e-satisfaction, providing quantitative evidence for e-commerce platforms to optimize the “she-design” field (Chou et al., 2015). The latest research exemplifies a strong trend toward data-driven and quantitative operations. Using a dynamic game model, it provides a quantifiable operational framework for synergistically optimizing online review incentive strategies and pricing strategies, reflecting the industry's deep exploration of the commercial value of user-generated content (Yang & Dong, 2018). Overall, this cluster of research closely tracks the e-commerce industry's evolution from traffic competition to “retention” operations, and from extensive growth to refined cultivation. Its methodologies uniformly employ quantitative analysis, highlighting the field's strong emphasis on data-driven decision-making. Future studies could explore the universality and particularity of these driving mechanisms by incorporating regional cultural differences.

In the field of digital consumption research, this cluster's studies exhibit a clear trajectory of theoretical deepening and technology-driven evolution spanning from 2008 to 2021. This closely mirrors the industry development path of e-commerce—from Web 2.0 socialization and information transparency to AI-powered personalized decision-making, as shown in Table 6. Early research established the theoretical foundation for core drivers. A study on the digital micro-product market confirmed that electronic word-of-mouth (e-WOM) aggregates multidimensional signals into “social proof” within homogeneous markets, serving as a key driver of sales. This provides empirical evidence for e-commerce platforms to build reputation marketing systems during their initial social transformation phase (Amblee & Bui, 2011). That same year, another pioneering study prospectively argued that the “consumer informatization” enabled by the internet was revolutionizing consumption behavior paradigms, giving rise to “long-tail” personalized consumption that transcends traditional mass markets. This profoundly revealed how enhanced information transparency—a fundamental industry driver—reshapes market structures (Lei et al., 2021).

As the platform economy matured, the focus of research shifted toward quantitative evaluation and exploration of motivation. To address the complexity of platform selection, advanced decision models, such as

Table 6
The top five authors of cluster 4.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
Harnessing the influence of social proof in online shopping: the effect of electronic word of mouth on sales of digital microproducts	Amblee and Bui (2011)	International Journal of Electronic Commerce	317	Quantitative analysis
Consumer informedness and diverse consumer purchasing behaviors: traditional mass-market, trading down, and trading out into the long tail	Clemons (2008)	Electronic Commerce Research and Applications,	64	Qualitative analysis
Model-based evaluation for online shopping platform with probabilistic double hierarchy linguistic codas method	Lei et al. (2021)	International Journal of Intelligent Systems	57	Quantitative analysis
An empirical study of information contribution to online feedback systems: a motivation perspective	Tong et al. (2013)	Information & Management	51	Quantitative analysis
Modeling personalized cognition of customers in online shopping	Wu and Liao (2021)	Omega-International Journal of Management Science	42	Quantitative analysis

the probabilistic dual-layer language CODAS, were introduced. By handling linguistic ambiguity, these models enhanced evaluation robustness, reflecting the demand for precise quantitative tools in the era of big data (Tong et al., 2013). Concurrently, empirical studies on motivations for participating in online feedback systems have revealed complex constraints involving altruism, self-interest, cost perception, and other multidimensional factors, providing a deep psychological foundation for designing effective user incentive systems (Wu & Liao, 2021). The latest research exemplifies the frontier trend of deep AI-driven empowerment. By constructing a multi-criteria decision framework integrating textual sentiment and quantitative parameters, the study achieves quantitative modeling of personalized consumer cognition. This marks a paradigm shift from group behavior analysis to precise individual insights, providing next-generation technical solutions for personalized recommendations and marketing automation on e-commerce platforms. In summary, this cluster of research began by validating the foundational value of e-WOM, progressed through theoretical interpretations of shifting consumption paradigms, advanced to platform evaluation and motivation mining using complex models, and ultimately reached personalized cognitive computing. This evolutionary trajectory profoundly reflects the development drivers of the e-commerce industry—from socialization and informatization to intelligentization—highlighting how academic research directly responds to technological transformation and commercial practice. Future research may further explore the applicability of these models and mechanisms across diverse regional market contexts.

At the intersection of digital consumption and health governance, relevant research has shown a clear evolutionary trend from basic information management toward comprehensive governance and precision interventions, as illustrated in Table 7. This trend profoundly reflects the core demands of e-commerce industry development: in its early stages (beginning in 2012), the focus was on addressing trust and integrity issues—such as fake reviews—through technological means to establish a secure transaction environment; As platform ecosystems matured, research shifted toward enhancing information decision-making efficiency (post-2017), helping users filter high-value content; more recently (2020), it expanded to leveraging e-commerce interfaces for health behavior guidance, demonstrating the growing social governance role of digital platforms. Specifically, at the forefront of health behavior interventions, a randomized controlled trial targeting food-insecure populations demonstrated that the "nudge" strategy of pre-populating nutritionally balanced shopping carts (default options) significantly increased the rates of healthy food purchases compared to traditional education. This provides empirical evidence for low-cost, scalable health interventions that utilize behavioral economics tools for specific groups, aligning with emerging industry trends that integrate public health with e-commerce fulfillment services (Coffino et al., 2020).

Regarding e-commerce anti-fraud technology, research has evolved from individual identification to group detection. Early studies constructed a heterogeneous graph model of "reviewer-review-store," overcoming limitations of traditional text analysis to achieve precise identification of highly covert fake review armies (Wang et al., 2012). Subsequent research introduced the Group Detection Algorithm (GSDB)

Table 7
The top five authors of cluster 5.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
Nudging while online grocery shopping: a randomized feasibility trial to enhance nutrition in individuals with food insecurity	Coffino et al. (2020)	Appetite	39	Quantitative analysis
Identify online store review spammers via social review graph	Wang et al. (2012)	Acm Transactions on Intelligent Systems and Technology	67	Quantitative analysis
Understanding the determinants of online review helpfulness: a meta-analytic investigation	Hong et al. (2017)	Decision Support Systems	325	Meta-analysis approach
A burst-based unsupervised method for detecting review spammer groups	Ji et al. (2020)	Information Sciences	43	Quantitative analysis
An analysis of review content and reviewer variables that contribute to review helpfulness	Malik and Hussain (2018)	Information Processing & Management	86	Quantitative analysis

based on reviewing outbreak patterns. By integrating kernel density estimation with behavioral feature analysis, it significantly enhanced defenses against organized review manipulation, reflecting a shift in platform governance strategies from reactive responses to proactive early warning systems (Ji et al., 2020). Regarding information quality assessment frameworks, a meta-analysis synthesized multiple studies to reveal common influencing factors, such as comment depth and timeliness. It systematically explained evaluation variations across platforms and product types, establishing a benchmark framework for the field (Hong et al., 2017). Further research leveraged machine learning models to integrate novel textual features, such as syllable count and space count, with reviewer historical data, significantly improving prediction accuracy for comment usefulness. This provides direct technical support for optimizing platform content ranking and display mechanisms (Malik & Hussain, 2018). In summary, this cluster of research systematically advances the development of a healthy, trustworthy digital consumption environment—from safeguarding environmental integrity and optimizing information quality to empowering positive consumer choices. Its evolutionary trajectory aligns closely with the e-commerce industry's development drivers: progressing from addressing foundational trust issues to enhancing user experience, and ultimately embracing broader societal value.

In the field of e-commerce consumer decision-making research, a clear trend has emerged: studies have progressively expanded from focusing on individual cognitive processes to examining social environmental influences, and ultimately integrating macro-socioeconomic effects. This evolutionary trajectory aligns closely with the development stages of the e-commerce industry—transitioning from a core focus on “information display” to “socialization and content-driven” models, and finally shifting attention to its external impacts, as shown in Table 8. Early studies, grounded in information processing and unconscious thinking theories, revealed that guiding consumers toward unconscious thinking patterns in high-information-load scenarios effectively enhances decision satisfaction. This finding provided theoretical support for e-commerce platforms to optimize user information processing pathways through interface design, reflecting the industry's early explorations in refining user experience operations (Gao et al., 2012). As social media and e-commerce became deeply integrated, research began examining the influence of emerging content formats.

Experimental comparisons revealed that, under specific market conditions, influencer recommendations significantly drive purchase intent more effectively than traditional user reviews. This conclusion highlights the “influencer economy” as a major driver of the industry, prompting platforms to reassess their content marketing resource allocation strategies (Dwidienawati et al., 2020). To address information overload, studies introduced the information foraging theory, experimentally demonstrating that consumers engage in phased, demand-driven “foraging” for different types of reviews. This research provides a direct theoretical model and technical pathway for platforms to achieve dynamic, personalized review information delivery, representing the evolution of e-commerce platforms from “information dumping” to “intelligent shopping guidance” (Li et al., 2017). Finally, through a systematic review that synthesizes years of research, it identifies the complex, bidirectional impact of B2C e-commerce on traffic mobility and critically notes that methodological limitations in prior studies have led to divergent conclusions. This macro perspective extends the influence of e-commerce decision research to urban governance and sustainable development, addressing the social responsibility issues that must be confronted as the industry matures (Rotem Mindali & Weltevreden, 2013). In summary, this cluster of studies comprehensively charts the evolution of e-commerce decision research from micro-level cognition to macro-level impacts, with its developmental momentum directly stemming from the core challenges and opportunities faced by the e-commerce industry at different stages (Table 9).

In the field of user technological immersion and e-commerce consumption behavior research, studies have demonstrated a trend toward deeper exploration of foundational theory development and specific behavioral mechanisms. This trajectory aligns closely with the industry evolution of e-commerce platforms, shifting from functional implementation to the deep optimization of user experience, enhanced user retention, and improved monetization efficiency. Groundbreaking work in this domain began with the systematic definition of the concept of “user immersion.” By integrating multidisciplinary perspectives, researchers constructed a theoretical framework comprising four dynamic stages and identified three core driving dimensions: users, systems, and interactions. This framework provides a crucial theoretical foundation and evaluation basis for subsequent quantitative studies on user experience, particularly regarding sustained engagement behaviors in e-commerce contexts (O'Brien & Toms, 2008). As online consumption became widespread, research began examining its potential negative impacts. A theoretical review first systematically proposed a conceptual model and predictive variables for “online shopping addiction,” introducing clinical psychology perspectives into consumer behavior studies. This addressed the industry's emerging need to identify and manage irrational consumption during its rapid growth phase (Rose & Dhandayudham, 2014).

Research further delved into specific moments of consumer decision-making. Studies based on large-scale user data revealed that the utilitarian and hedonic values of online reviews stimulate browsing behavior, thereby triggering impulse purchases, with individual impulsive traits moderating this process. This finding illuminates the psychological mechanism by which content marketing drives immediate conversions through a “value-to-behavior” pathway, offering direct insights for platforms to optimize review displays to stimulate sales (Zhang et al., 2018). Concurrently, another study focused on cross-category recommendation strategies commonly employed by platforms, identifying decision difficulty as a key moderating variable. This suggests that effective recommendation strategies require dynamic adjustments based on users' real-time decision states, reflecting the technology-driven evolution of e-commerce operations from “one-size-fits-all” broad recommendations toward “context-sensitive” precision targeting (Zhu et al., 2018). In summary, this cluster of studies comprehensively outlines how e-commerce platforms, starting from understanding the fundamental mechanisms of user immersion, have progressively

Table 8
The top five authors of cluster 6.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
Customer review or influencer endorsement: which one influences purchase intention more?	Dwidienawati et al. (2020)	Heliyon	43	Quantitative analysis
Understanding online purchase decision making: the effects of unconscious thought, information quality, and information quantity	Gao et al. (2012)	Decision Support Systems	135	Quantitative analysis
Sequentially of product review information provision: an information foraging perspective	Li et al. (2017)	Mis Quarterly	36	Quantitative analysis, qualitative analysis
Transport effects of e-commerce: what can be learned after years of research?	Rotem Mindali and Weltevreden (2013)	Transportation	108	Qualitative analysis

Table 9
The top five authors of cluster 7.

Article	Authors	Journal	Total citations (in WoS Core Collection)	Methodology
What is user engagement? A conceptual framework for defining user engagement with technology	O'Brien and Toms (2008)	Journal of the American Society for Information Science and Technology	43	Conducted semistructured interviews
Towards an understanding of internet-based problem shopping behavior: the concept of online shopping addiction and its proposed predictors	Rose and Dhandayudham (2014)	Journal of Behavioral Addictions	106	Theoretical review of the literature
Online reviews and impulse buying behavior: the role of browsing and impulsiveness	Zhang et al. (2018)	Internet Research	125	Quantitative analysis
The influence of online cross-recommendation on consumers' instant cross-buying intention: the moderating role of decision-making difficulty	Zhu et al. (2018)	Internet Research	30	Quantitative analysis

developed systematic strategies capable of guiding or even shaping specific consumer behaviors (including potentially risky ones). This profoundly reflects the industry's intrinsic logic of achieving growth through deep insights into user psychology.

3.3. Discussion: integrating findings and theoretical dialogue

Fig. 9 illustrates the framework of the "Multi-dimensional Data-Driven Decision-Making Ecosystem for Online Consumer Behavior" constructed in this study. Based on the seven research clusters identified through the bibliometric analysis in Section 3.2, the selection of its constituent elements and the design of its associative pathways both stem from content analysis of the empirical clustering results, establishing a clear literature foundation. The framework reveals the multi-layered, dynamic mechanism through which online reviews influence consumer purchasing decisions.

Specifically, the system begins at the Data Governance and Trust Foundation layer (Cluster 5: Fake Review Governance), ensuring the information quality and credibility of the review ecosystem. Building upon this foundation, the Technology Empowerment and Information Processing layer (Cluster 1: Sentiment Analysis; Cluster 2: Technology Empowerment; Cluster 4: Behavioral Modeling) utilizes advanced tools such as natural language processing, artificial intelligence, and big data modeling to extract review value and influence consumer perceptions. Processed information then flows into the Behavioral Conversion and Decision Layer (Cluster 3: Word-of-Mouth and Loyalty; Cluster 6: Decision Mechanisms; Cluster 7: Psychological Intervention). At this level, social interactions (e.g., word-of-mouth transmission), individual cognitive mechanisms (e.g., decision trade-offs), and psychological factors (e.g., impulse regulation) collectively drive purchasing behavior. Cultural context and individual differences, represented as dashed lines in the diagram, serve as crucial moderating variables that permeate and influence all layers.

The association paths between clusters (indicated by arrows in the diagram) and their assigned influence strengths (represented by arrow line types) are not arbitrary assumptions. Instead, they are derived from content analysis of highly cited literature within each cluster, identifying primary (strong influence, solid lines) and secondary (weak influence, dashed lines) theoretical relationships. For instance, the driving role of sentiment analysis (Cluster 1) as a technological foundation for trust building (Cluster 2) and word-of-mouth communication (Cluster 3) has been repeatedly validated in numerous empirical studies, hence its connection via solid lines.

The theoretical contribution of this framework lies in integrating fragmented factors identified through bibliometric analysis into an internally coherent "technology-society-psychology" system. The

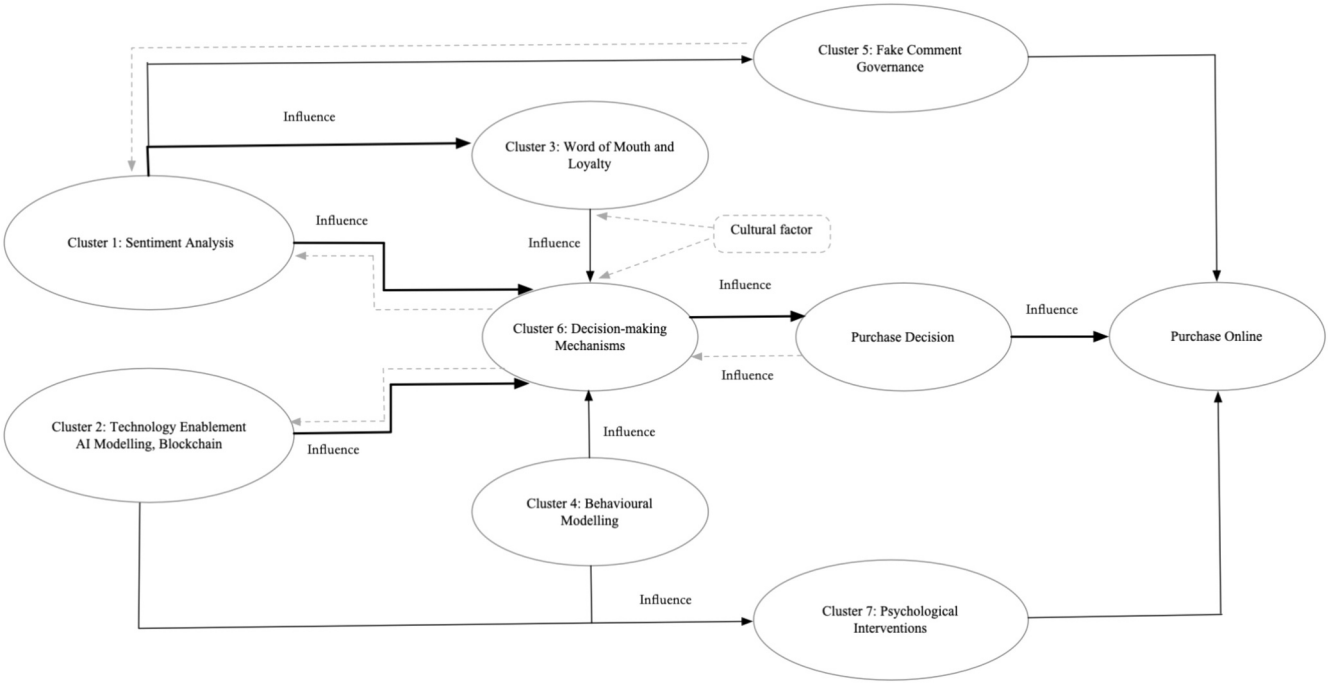


Fig. 9. Multidimensional data-driven decision-making ecosystem (▬ denotes strong impact, ▬ denotes weak impact).

framework's construction follows a logical progression from empirical data (metacommunication clustering) to theoretical induction (content analysis), with its structure supported by existing theoretical clusters. For instance, Information Integration Theory focuses on how individuals' weight multi-attribute information, while this framework further reveals how social interaction (Cluster 3) and technological platforms (Clusters 2 and 4) act as external agents reshaping this weighting process. The Theory of Planned Behavior emphasizes antecedents of behavioral intent, while this framework elaborates how online reviews collectively alter intent by influencing consumers' attitudes (cognitive level), subjective norms (social level, derived from Cluster 3), and perceived behavioral control (affected by Cluster 2's platform trust mechanisms).

Furthermore, the framework implicitly incorporates key stakeholder roles: consumers are the decision-making core, whose cognitive and emotional responses are shaped by review information and platform technology; Sellers's attempt to influence decisions by generating information and engaging in word-of-mouth interactions; E-commerce platforms act as intermediaries, regulating information flows and interaction patterns through algorithmic and mechanism design. This indicates that the study's findings not only synthesize existing knowledge but also provide a more integrated and empirically grounded theoretical perspective by clarifying systemic structures and interactive relationships.

4. Open issues, challenges, and future research directions

4.1. Open issues and challenges

4.1.1. Open problems and challenges I: bottlenecks in generalization capabilities for cross-platform and cross-cultural sentiment analysis

This challenge stems directly from the limitations exhibited by Cluster 1 (sentiment analysis) when handling multimodal, cross-cultural review data, as well as the moderating effects of cultural and individual differences emphasized by Clusters 6 and 7. Current sentiment analysis techniques (e.g., SLCABG, LSIBA-ENN) perform well in a single platform or language environment. Still, when facing multi-platform (e.g., e-commerce, social media, live streaming) and cross-cultural (e.g., semantic differences between Chinese and English comments, differences in the cultural expression of sentiment) environments, the model's generalization ability is significantly limited. In dynamic scenarios, the real-time, multimodal (text + emoji + video) characteristics of comments further exacerbate the risk of sentiment polarity misjudgment (Brand & Reith, 2022). Additionally, there are cultural expression differences in various cultural contexts. Based on Hofstede's theory of cultural dimensions (Hofstede, 1980), consumers may express opposite emotions for the same product (e.g., differences in the praise and criticism of 'cost-effective'), such as Chinese consumers often use implicit expressions (e.g., 'okay' is actually a bad review), while Western users tend to directly evaluate negatively (e.g., 'not bad' is actually a bad review). At the same time, there are semantic decoding biases. In high-context cultures, emotions are context-dependent (e.g., 'good' in Chinese needs to be combined with ratings), resulting in a higher model misclassification rate than in English (Qi & Jin, 2024). As a result, it is challenging for existing models trained on large-scale corpora to capture nuanced cross-cultural sentiments. Breaking through platform and cultural boundaries to build a universal sentiment analysis framework that can adapt to multimodal data, dynamic scenarios, and cultural differences has become an urgent challenge.

4.1.2. Open issues and challenges II: the dilemma of reconstructing trust mechanisms in dynamic decision-making environments

This challenge arises from analyses within Cluster 2 (Trust Building) and Cluster 5 (False Review Governance), particularly as traditional trust mechanisms face increasing challenges in emerging dynamic scenarios such as live-streaming e-commerce. Traditional trust construction mechanisms (e.g., review systems, seller guarantees) have verified effectiveness in static scenarios, but face the risk of failure in dynamic decision-making environments such as live e-commerce and cross-border shopping. For example, the instant interaction of anchors in live streaming scenarios may weaken consumers' reliance on historical reviews. At the same time, the weights of institutional trust (e.g., platform regulation) and capability trust (e.g., logistics efficiency) in cross-border shopping may change due to cultural differences and policy fluctuations. In addition, information overload and false information (e.g., mercenary reviews, false advertisements) further blur the boundary of trust, leading to a shift from 'rational assessment' to 'intuitive judgement' in the decision-making path of consumers. How to quantify the impact of dynamic trust elements (e.g., real-time interactions, group behaviors) on decision-making and designing a trust assessment model that can be adjusted in real-time is the key to improving transaction conversion rates and user retention.

4.1.3. Open issues and challenges III: ethical boundaries and sustainability contradictions of behavioral intervention techniques

This challenge is closely related to studies in Cluster 7 (Psychological Intervention) and Cluster 4 (Behavioral Modeling), which, while demonstrating the effectiveness of interventions, also reveal potential threats to ethics and user autonomy. Although behavioral intervention technologies (e.g., default options, personalized recommendations) can effectively improve purchasing rates or decision-making efficiency, their potential ethical risks (e.g., deprivation of user autonomy, privacy leakage, information cocooning) have triggered extensive controversies (Karthik & Ganapathy, 2021). For example, although pre-populated shopping carts can guide healthy consumption, they may weaken users' active learning of knowledge. Although personalized recommendations can enhance user experience, they may exacerbate information bias and consumption alienation. In addition, there is a contradiction between the short-term effects of intervention strategies and their long-term sustainability: over-reliance on behavioral interventions may lead to users' dependence on technology, weakening their autonomous decision-making ability. Finding a balance between the effectiveness of behavioral interventions and user autonomy and privacy protection is a core challenge for the healthy development of the digital consumption ecosystem.

4.2. Future research directions

4.2.1. Direction I: cross-modal semantic fusion and dynamic adaptive sentiment analysis

This direction corresponds to Challenge I, stemming from the technological frontier of Cluster 1 and the regulatory variables of Clusters 6 and 7. Future research needs to overcome the limitations of single modality (text) and static corpora, and build a framework for sentiment analysis that integrates multimodal data (text, image, speech, video) and dynamic semantic networks. For example, the correlation of multimodal elements in comments (e.g., the synergistic expression of textual emotions and emoticons) and their evolving trends are captured by introducing graph neural networks (GNNs) with spatio-temporal attention mechanisms (Xu et al., 2024). Meanwhile, a cross-platform migration model based on Meta-Learning (ML) is developed to quickly adapt to new scenarios using a small amount of domain data and dynamically adjust the model parameters to cope with semantic drift through reinforcement learning. In addition, we explore the cross-cultural emotion lexicon and culturally adaptive pre-training model embedding IDV/UAI feature vectors to regulate the model's attention, quantify the impact of cultural differences on emotion expression, and adopt cross-cultural adversarial training to reduce contextual ambiguity interference to enhance the model's generalization ability in globalized scenarios, according to the needs. Furthermore, based on the need to explore cross-cultural sentiment dictionaries and culturally adaptive pre-training models, we adjust model attention by incorporating IDV/UAI features,

quantify the impact of cultural differences on emotional expression, and employ cross-cultural adversarial training to reduce contextual ambiguity interference and enhance the model's generalization capability in globalized scenarios.

4.2.2. Direction II: dynamic trust network and multi-subject game strategy optimization

This direction corresponds to Challenge II, stemming from the research foundations of Clusters 2 and 5. to meet the demand for trust reconstruction in dynamic decision-making environments, future research can build a dynamic trust network based on blockchain and graph neural network, incorporating users, merchants, platforms, KOLs and other subjects into a unified trust framework, updating trust weights in real time, and tracing back the source of trust. For example, through the introduction of Federated Learning and differential privacy technology, the collaborative computation of multi-object trust data can be achieved under the premise of data security. At the same time, game theory and reinforcement learning are combined to simulate the strategic interactions of multiple subjects in trust construction (e.g., how merchants can enhance trust through dynamic pricing and guarantee strategies, and how users can reverse supervise the platform through group behaviors), so as to optimize the allocation of trust resources and the risk prevention and control mechanism (Ding et al., 2025). Additionally, a cross-cultural trust assessment model needs to be developed to quantify the impact of cultural values (e.g., collectivism vs. individualism) on the trust mechanism and to provide differentiated trust strategies for globalized e-commerce platforms.

4.2.3. Direction III: ethics-driven behavioral intervention and user digital literacy enhancement

This direction corresponds to Challenge III, originating from Clusters 4 and 7, and addresses ethical concerns. Future research should focus on establishing ethical norms, designing transparent and explainable behavioral intervention strategies, and ensuring the sustainability of intervention techniques through user empowerment. For example, developing intervention tools based on Choice Architecture (Wang et al., 2020), embedding “opt-out mechanisms” and “self-selection prompts” within default options to safeguard user autonomy (Parikh & Parikh, 2017); leveraging blockchain technology—specifically smart contracts and zero-knowledge proofs—to build decentralized trust and evaluation systems, ensuring user data sovereignty and privacy security. For instance, Taobao's “Privacy Protection Umbrella” initiative allows users to generate virtual interest profiles for recommendation purposes while storing real data locally. Simultaneously, introduce a “digital literacy” assessment framework. Utilize interactive educational tools (e.g., virtual experiments, decision simulations) to help users recognize and resist excessive manipulation (e.g., information silos in personalized recommendations). For instance, on Amazon, consumers can use the “Algorithm Transparency Lab” to simulate how recommendation algorithms work, adjust parameters to observe outcome changes, and thereby understand information filtering mechanisms (Wang et al., 2021). Additionally, an ethical assessment system for behavioral interventions is needed to quantify the short-term effects and long-term social impacts of intervention strategies, thereby providing a scientific basis for policy formulation.

5. Conclusions

This study employs a systematic literature review and bibliometric analysis to deeply uncover the complex influence mechanisms of online reviews on consumer purchasing decisions. It systematically distills seven key thematic clusters: sentiment analysis, online trust building, word-of-mouth propagation, consumer behavior modeling, fake review identification, decision mechanisms, and user psychological intervention. The study first systematically identified the key factors by which online reviews influence consumer decisions (addressing RQ1). Subsequently, by constructing a “multi-dimensional data-driven decision ecosystem” framework, it elucidated the multi-level interactive mechanisms among these factors (addressing RQ2). Findings reveal that consumer online shopping decisions constitute a multidimensional dynamic process. They are not only directly influenced by the emotional orientation and content credibility of review texts but also involve deep-seated interactions between user psychological mechanisms and behavioral decision models. This framework offers an integrated perspective for understanding consumer behavior in the digital age. It reveals the objective value of online reviews as an information medium while emphasizing the central role of consumers' subjective cognition and emotional connections in building trust and driving loyalty, thereby addressing existing research gaps in interdisciplinary theoretical integration. The seven major research clusters identified by bibliometric analysis, along with the three open challenges and future directions derived from them, suggest that this field is transitioning from single-factor analysis to integrated research on complex systems.

At the practical level, the research results suggest a straightforward guiding path for the e-commerce industry. By efficiently integrating natural language processing and machine learning technologies, enterprises can accurately capture user preferences from massive reviews, dynamically optimize product and service design, and predict market trends. At the same time, the intelligent transformation of the platform interface and the deepening of the application of personalized recommendation systems can not only improve the efficiency of consumer decision-making but also help build a long-term, stable user relationship network. It is worth noting that the breakthrough in false review identification technology and the development of consumer privacy protection mechanisms are becoming key propositions for balancing data value mining and user rights protection, which will directly impact the sustainable development of the future e-commerce ecosystem.

This study has the following limitations. First, at the methodological level, the bibliometric analysis primarily relies on quantitative features from the Web of Science database. It is constrained by search strategies and database coverage (e.g., not systematically incorporating platforms such as Scopus and Dimensions), potentially failing to capture all relevant literature in the field. Furthermore, its ability to analyze subtle contextual differences in comments and the qualitative characteristics of cross-cultural consumption behaviors is limited. Second, in terms of content coverage, existing analytical frameworks primarily focus on text-based comments, with insufficient research on user feedback mechanisms in emerging media formats such as short video comments and voice interactions.

Given these limitations, future research may expand in the following directions. Methodologically, incorporating large language models (e.g., generative pre-trained transformers) for the deep semantic deconstruction of comments, combined with reinforcement learning techniques to dynamically simulate consumer decision paths, can enhance the understanding of complex contexts and cross-cultural variations. At the data level, expanding the fusion analysis of multi-source heterogeneous data—including cross-platform social media unstructured comments and biosensor data—is recommended to more comprehensively capture consumer sentiment and behavioral responses. Regarding collaborative mechanisms, exploring partnerships with major e-commerce platforms (e.g., Amazon, Alibaba) to establish federated learning-based data collaboration frameworks could enable secure mining and integrated analysis of multi-source comment data while safeguarding user privacy. Furthermore, actively advancing blockchain technology applications in comment traceability and credibility assessment is essential. Building decentralized, tamper-proof comment evidence systems will enhance transparency and trustworthiness within the comment ecosystem. These directions hold promise for deepening the integration of theoretical development and methodological innovation in consumer behavior research, providing systematic solutions for precision marketing and platform governance in the digital era.

CRediT authorship contribution statement

Qunli Tang: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Investigation, Formal analysis, Data curation. **Peng Jiang:** Supervision, Software.

Ethics approval

This study is a systematic review of publicly available literature and does not involve direct interaction with human participants, animal subjects, or primary data collection. Therefore, ethics approval was not required.

Declaration of Generative AI and AI-assisted technologies in the writing process

We hereby declare that no generative artificial intelligence (AI) tools or technologies were utilized in the preparation, writing, or analysis of this manuscript. This work is entirely the product of the authors' own intellectual labor.

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